

SATYAM VIRAT

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PROFILE

Software Engineer with ~ 4 years of experience designing and building scalable backend systems, distributed microservices, and production AI platforms using **Python, FastAPI, Node.js, Java, and AWS**. Experienced in **GenAI, LLMs, RAG, AI agents, event-driven architectures, Kafka, Redis, WebSockets, and real-time systems**, with hands-on expertise across **PostgreSQL, MySQL, MongoDB, Docker, and CI/CD**. Strong focus on system design, performance optimization, security, scalability, observability, and production reliability, with experience delivering high-throughput backend services and architecting scalable polyglot cloud platforms.

EXPERIENCE

Backend-AI Engineer (FastAPI, GenAI, RAG, Docker, AWS, MongoDB, Celery) Feb 2026 - Present
ORVERA AI *Pune, Maharashtra, India*

- Architected a production-grade AI Voice/Chat Agent using FastAPI, Celery, Redis, WebSockets & VoIP supporting 300+ concurrent live calls/sessions with streaming LLM responses.
- Orchestrated a sophisticated **hybrid RAG** pipeline (BM25, FAISS and FlashRank) linking LLMs with STT/TTS and tool-calling capabilities, enhancing real-time conversational AI and knowledge retrieval accuracy up to 95%.
- Standardized **MongoDB** by migrating from synchronous MongoEngine/Motor to PyMongo Async, eliminating thread-pool overhead and simplifying async data access across 400+ APIs and 30+ Celery tasks.
- Led Docker deployments via GitHub Actions, AWS ECR & ECS, reducing manual deployment time by 25 mins.
- Improved AI safety and platform security through prompt guardrails, VAPT remediation, AWS, KMS/Secrets Manager, **achieving 90% PCI-DSS compliance**.
- Leveraged AI coding agent Codex to accelerate feature delivery by 3× while maintaining production quality.

SDE - 1 (Python, AWS, PostgreSQL, MySQL, Redis, CI/CD, Kafka) Jun 2024 - Jan 2026
1SILVERBULLET *Pune, Maharashtra, India*

- Delivered 12 production-grade insurance products, building scalable **distributed microservices** with REST APIs, webhooks, and background jobs while optimizing MySQL/Redis to reduce API latency by 40%+.
- Constructed an **AI-powered** Quality Management pipeline using Python, AWS & **LLMs** to batch-analyze 5K+ interactions in 1 hour, assessing agent performance, policy adherence, drop-offs & generating actionable remarks.
- Implemented RBAC, SSO, and Kafka pub/sub workflows to strengthen platform security and reliability.
- Reduced MTTR by 70% by implementing centralized logging, tracing & observability with AWS CloudWatch.
- Automated **CI/CD** with Bitbucket Pipelines, CodeDeploy, Docker & ECS, cutting deployment time by 90%.
- Contributed to architecture design, production rollouts, PR reviews, and mentored engineers on SOLID, OOPS & design patterns which decreased production defects by 18% and boosted maintainability.
- Eliminated duplicate payment scenarios via **idempotent payment** handling; enabled a client to **achieve 1,000+ policies** in 3 months — the fastest adoption across all live products.

MERN Full Stack Developer Trainee (Node, React, MongoDB, Express.js) Jan 2023 - May 2024
Coding Ninjas *Remote, India*

- Spearheaded a full-stack job-hunting platform supporting **3 core domains** - users, jobs, and applications by architecting React, Node.js/Express, and MongoDB components around modular REST APIs.

- Engineered 15+ **REST APIs** for user profiles, job discovery, job posting, and application management utilizing structured Express.js controllers, MongoDB data models, request validation, and centralized error handling.
- Implemented secure authentication and authorization across 3+ protected user roles by integrating **OAuth 2.0 for identity authentication and JWT-based access tokens** for stateless API authorization and protected resource access.
- Developed 10+ **responsive React views** covering authentication, job search, job details, profiles, and application tracking by building reusable components and integrating them with backend REST services.
- Containerized and deployed the complete application stack on **AWS EC2** by Dockerizing frontend/backend services, configuring **Nginx as a reverse proxy**, and integrating **MongoDB Atlas** for managed cloud database connectivity.

PROJECTS

Visual Workflow Editor

CareerOS-Your Career Buddy — AI-Powered Career Platform

- Architected an AI-powered career platform across **8+ domain services** covering identity, profile intelligence, job discovery, community collaboration, notifications, AI intelligence, application tracking, and interview workflows using a progressive **polyglot microservices architecture**.
- Pioneered service boundaries across **3 backend technology domains** — Node.js/Express for platform and integrations, Python/FastAPI for AI intelligence, and Java/Spring Boot for transactional application workflows — enabling independent evolution and scaling of domain services.
- Engineered a multi-database architecture using **4 data stores** — MongoDB, PostgreSQL, Redis, and object storage S3 — by enforcing strict service-level data ownership and selecting persistence models according to workload, consistency, and access-pattern requirements.
- Streamlined an event-driven communication layer across **5+ domain events**, including job-source monitoring, resume analysis, application creation, notifications, real-time updates, using **Kafka, Redis, WebSockets and Celery** for asynchronous processing while retaining REST for synchronous operations.
- Built Cady, a personalized AI career assistant, across **6+ intelligence capabilities** including resume/JD analysis, skill-gap detection, ATS analysis, job matching, recommendations, and company-specific preparation using **LLMs, RAG, vector embeddings, and NLP/ML pipelines**.
- Established a progressive cloud deployment strategy by containerizing services with **Docker** and separating application workloads into independently deployable services, enabling a structured evolution toward **AWS, CI/CD, observability, and horizontally scalable production infrastructure**.

SKILLS

AI/ML	GenAI, LLMs, RAG, FAISS Vector DB, BM25, Prompt Engineering, AI Agents, Embeddings
Backend	Python, FastAPI, Node.js, Kafka, REST APIs, Microservices, Distributed Systems
Frontend	React, JavaScript, HTML, CSS
Databases	PostgreSQL, MySQL, MongoDB, Schema Design, DB Migrations, Query Optimization
Infra & Cloud	AWS (EC2, ECS, S3, DynamoDB, CloudWatch, IAM, KMS, Secrets Manager), Redis, Celery
Tools & Practices	Git, Linux, CI/CD, System Design, OOP, SOLID, Role-based Authentication/Authorization

EDUCATION

THDC-IHET, Uttarakhand Technical University

Bachelor of Technology in Computer Science Engineering, CGPA 7.6

AWARDS AND CERTIFICATIONS

- Honored with **5th PI Star** at 1SilverBullet for outstanding performance.